UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

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The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

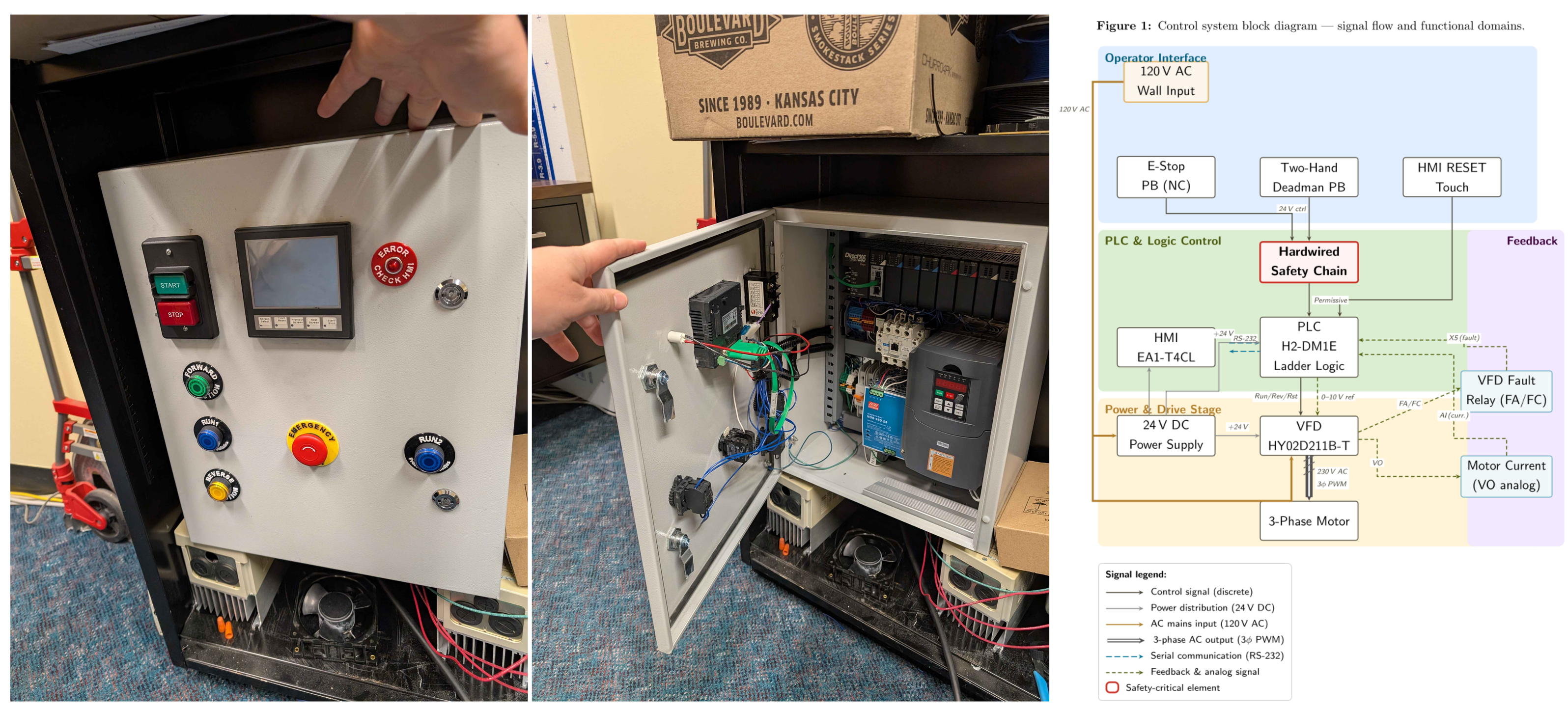
Must:

- Shred plastic to ∼1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the $1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.
May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hard-wired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 2 HP IronHorse MTCP2-002 premium-efficiency TEFC 3-phase induction motor (230 V, 5.93 A FLA, 1735 RPM, 145TC frame, VFD-rated 1.0 SF) over U/V/W via blue, white, and red 4 mm banana sockets.



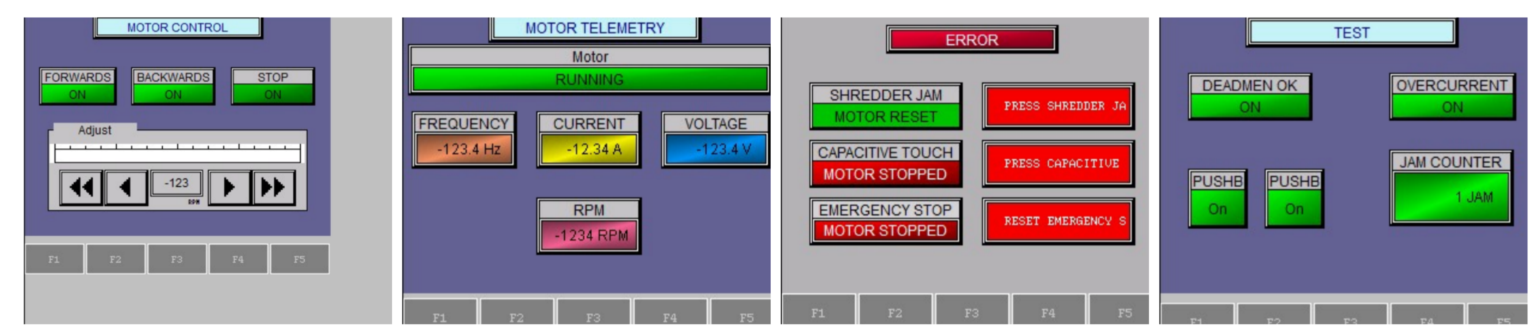
Operator face (left), panel interior (middle), and system block diagram (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks; block diagram shows the signal path from AC in through the safety chain to the VFD and motor.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: `D2-08ND3` (8 DC inputs), `F2-08AD-1` (analog current input), `F2-04RTD` (RTD), `D2-08TR` (8 relay outputs), and `F2-08DA-2` (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

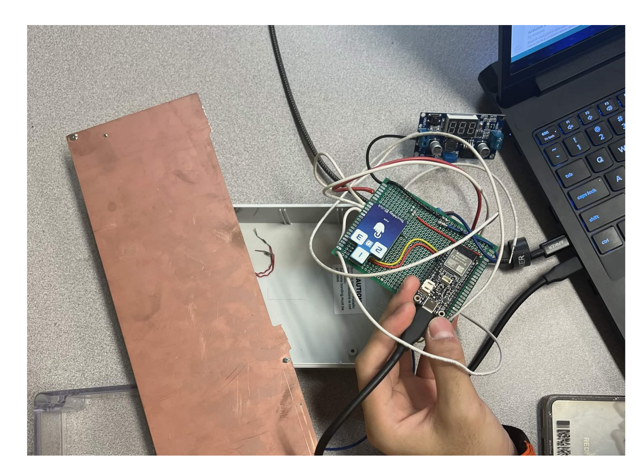
AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running `HMI_413.mgp`, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control • Motor Telemetry • Error / Alarm • Test.

Capacitive-Touch Safety (CTSI) Module

A contact-free operator-presence sensor designed and built in-house by the Shredder EE team. An ESP32 reads a capacitive electrode; on a touch event the ESP32 drives an LED through an optocoupler that closes the 24 V PLC input `X7`. Power is taken from the panel 24 VDC rail through an LM2596 buck converter to the ESP32's 3.3 V rail.

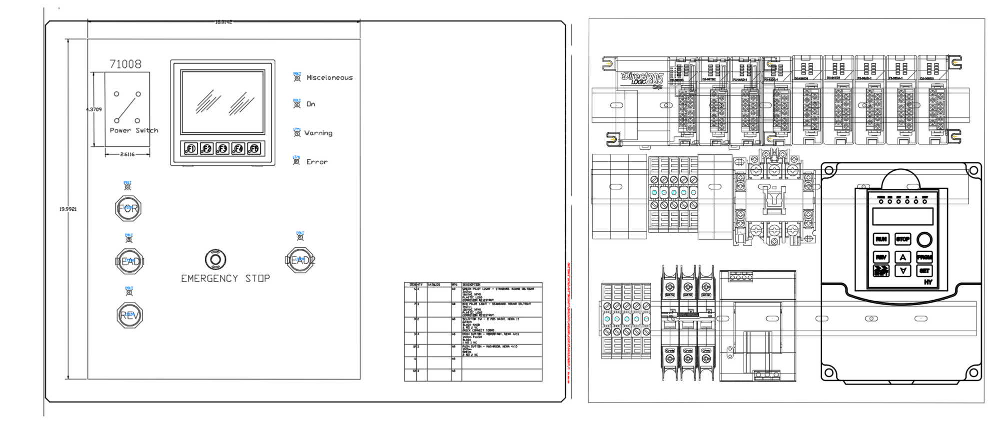


ESP32 + LM2596 buck + optocoupler module wired to PLC `X7`.

Ladder response (rung 28): touching the electrode latches `Y4` (fault LED), `C15` (`STOP_MOTOR`), `C21` (`CTSI_TRIG`), and `C22` (`CHANGE_SCREEN`) – motor stops, fault LED on, HMI jumps to ERROR, all independent of the deadman / E-stop chain.

Status: bench-validated against the PLC; full machine-level verification on the enclosed prototype pending.

Panel CAD Layout



AutoCAD Electrical: front operator face (left) with power switch, HMI cut-out, status lights, fwd / rev / deadman pushbuttons, and E-stop; back-plate component layout (right) with terminal blocks, DirectLOGIC 205 rack, SD-N35 contactor, fused 24 V branch, Huanyang VFD, and HMI cut-out.

Safety & Compliance

- **E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- **Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- **Overcurrent / jam:** VFD over-torque trip (`PD124 = 150%`, `PD125 = 3 s`) on PLC X5; ladder runs the 3-strike unjam routine.
- **Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- **Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B–1 / PL b–c), 13850, 13851, OSHA 1910.147.

Results

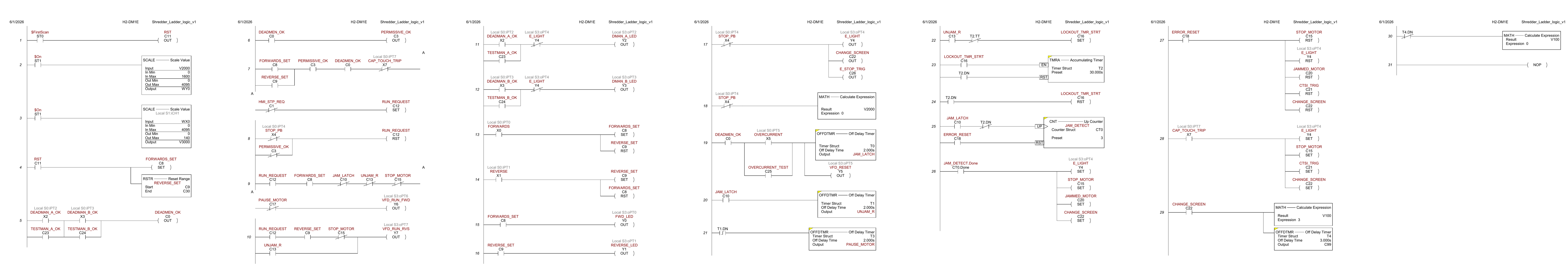
All EE "Must" requirements met for the items the EE team controls: VFD motor control, NFPA-70 design, E-stop + overload + safety interlock, HMI control & status, and the $1,000 budget.
Constraint: the ME team did not complete the mechanical shredder build by end of term, so full machine-level integration testing (loaded shredding of plastic samples on the actual frame) was not possible. All EE validation is bench-only against the VFD and motor stand-in; the panel is ready to integrate as soon as the mechanical prototype is delivered.
Working:

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: HMI `V2000` → `SCALE` → `F2-08DA-2 CH1` → VFD `VI`.
- Jam-recovery verified: overcurrent on `X5` → 2 s reverse (`T1`) → coast-to-stop buffer (`T3`).
- Three-revision point list signed off by Fearghus Tyler.

Wired and ready, programming pending: Everything below is fully wired into the panel and the ladder program; what's left is configuration / firmware work, not hardware.

- **CTSI module not yet integrated** – the capacitive-touch breakout and its microcontroller were destroyed during bench bring-up. The X7 input, optocoupler interface, and rung-28 ladder response are still in place and ready for the replacement module to drop in.
- **Jam-count telemetry not finished** – counter `CT0` / `N0` increments correctly inside the PLC but is not yet surfaced on the HMI for live readout.
- **Voltage / current telemetry not working** – plan was to take the VFD's frequency and motor-current analog outputs into the PLC and back-calculate the derived voltage, RPM, and status fields shown on Screen 2. The path VFD `VO` → F2-08AD-1 `CH1` → `WX0` → `V3000` is wired and rung 3 scales it, but VFD analog-output parameters and the PLC scaling still need to be revisited before the live values stabilize.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN_REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

31 ladder rungs in production.
27 HMI tags across 4 screens.
14 component sheets in the point list.
3 strikes before the motor locks out.
2 s unjam reverse stroke.
150% over-torque threshold on the VFD.
$969.88 spent against the **$1,000** control-panel budget (**$30.12** under).

Hardware Stack

PLC: DirectLOGIC 205 + H2-DM1E.
HMI: C-more EA1-T4CL.
VFD: Huanyang HY02D211B-T.
Motor: IronHorse MTCP2-002 (2 HP, 1735 RPM, 145TC, TEFC, VFD-rated).
PSU: Mean Well NDR-480-24.
Contactor: Mitsubishi SD-N35.

Future Work

Production motor selection (3–5 HP) supersedes `PD141--PD144`.
Mechanical integration and full shred-load run pending – ME team did not deliver the prototype by end of term.
Capacitive-touch (CTSI) module on `X7`: machine-level verification pending.
Modbus link to the VFD for richer telemetry (deferred – the Modbus module exceeded the $1,000 control-panel budget, so the analog `VO` / `VI` path was used instead).

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsoring the project; to advisors **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton**; to the **ME capstone team** for the mechanical design and ongoing integration coordination; and to the **PSU Power Engineering Laboratory** for bench access.

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